

Jason Hong

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EDUCATION

Massachusetts Institute of Technology

Cambridge, MA

B.S. in Computer Science and Engineering

Sept 2024 – May 2028

- GPA: 5.0/5.0
- Relevant Coursework: Operating Systems Engineering, Digital Systems Laboratory (FPGA), Computer Architecture, Data Structures and Algorithms, Fundamentals of Programming
- Leadership: Associate Advisor, MIT Chinese Students and Scholars Association

EXPERIENCE

Durability

Cambridge, MA

Software Engineering Intern

June 2026 – Aug 2026

- Built a full-stack React/Supabase coaching platform and ETL pipeline to import and normalize 48 legacy TrueCoach programs (7,600+ exercise prescriptions) into editable multiweek calendars supporting versioned publishing and athlete assignment.
- Built an end-to-end bulk roster-upload pipeline across React, Supabase/Deno, and SwiftUI, automating validation, batched invitations, email OTP, and first-login password setup.

Kyte Dynamics

Los Angeles, CA

Software Engineering Intern

May 2025 – Sept 2025

- Built custom bidirectional MAVLink/uORB messaging between two PX4 flight controllers over UART and validated it on hardware, implementing outbound streaming, inbound decoding, and test publication.

MIT 6.191 Computation Structures

Cambridge, MA

Lab Assistant

Sept 2026 - present

- Responsible for hosting office hours, tutoring students, and providing lab checkoffs.

RESEARCH

MIT Laboratory for Information & Decision Systems

Cambridge, MA

Undergraduate Researcher (UROP) | Advised by Prof. Tamara Broderick

April 2026 – Present

- Migrated [Nomon](#)'s language-model interface to the TextSlinger API to support language-model benchmarking.
- Corrected inconsistent conditioning of letter and word probabilities in log space, reducing the simulated correction rate from 3.99% to 0.68%.
- Developed output-validity checks that revealed 33.5% of word suggestions were invalid, a failure undetected by the oracle simulator; restricting suggestions to a valid vocabulary eliminated invalid outputs and reduced simulated click load from 1.390 to 1.147.

HONORS & AWARDS

MIT Pozen Fellowship: Awarded to select students for excellence in engineering and public service (2026)

Dr. Sherrie Emoto Mathematics Scholarship: Highest departmental honor for mathematical achievement at Fullerton College (2025)

PROJECTS

RISC-V Neural Network Inference Accelerator | Minispec/BSV, C, RISC-V

Spring 2026

- Optimized a pipelined processor and MNIST inference stack, achieving an **80×** improvement over baseline.
- Extended the ISA with custom instructions and pipelined SIMD multiplication; cut clock period from **1072 ps** to **529 ps** via a Kogge-Stone adder and by re-encoding ALU branch output as a single comparison bit, and added a specialized iCache to reduce CPI.

TECHNICAL SKILLS

Languages: C/C++, Python, React, Typescript Swift, Minispec, Bash

Tools & Frameworks: Linux/Unix, Git, Neovim, SwiftUI, React, Supabase, CMake, MAVLink

Interests: Computer Architecture, Verified Software Engineering, OS Kernels, Performance Engineering